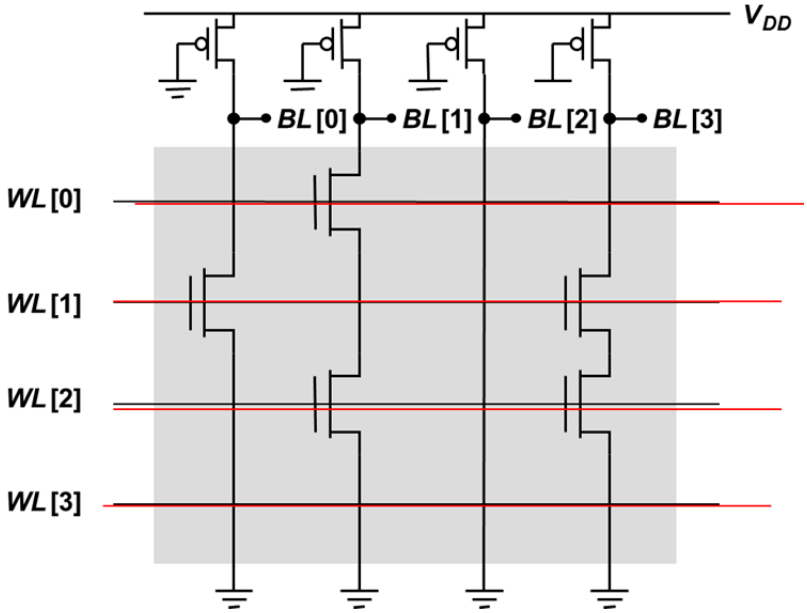
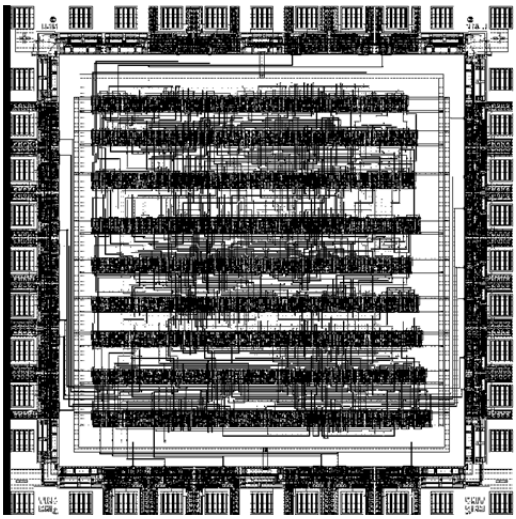
Final Exam. EC571 Fall 2012. FIVE pages Name_____

1. **What are the bits (zeros and ones) stored in this ROM (8 points). There are 16 answers.**



2.(6)A.What is the likely design methodology that made the chip below. B.Circle ONE pad



The sheet resistance is 10^4 ohms per square for a turned-on NMOS and use double that value for PMOS.

Cg=10-14 Farads for a transistor whose size is 2 units by 2 units. Cg is the capacitance of a 2 x 2 NMOS or PMOS gate. Use V_T = 1 V for nmos and –1V for pmos. Unless stated otherwise, assume Vdd=5V. Note: variables are given by single characters, e.g. A, B, …, and concatenated character strings such as AB are to be read as A and B. A' means Acomplement. A + B means A or B.

3) (10)Consider a fully complementary CMOS NAND(A,B,C) using three transistors in series to ground. Estimate pullup and pulldown (PU/PD) times for each input combination switching from high to low or low to high. Assume an ideal pulse source that drives the changes in A, B, and C. For each of the changes, fill in the table with estimated rise and fall times against a load that is 10**-11 F Assume the pullup transistors are 4w x 2long and the pulldowns are 2 x 2. Assume the sheet resistance of an "on" NMOS transistor is 10 kohms and double that for PMOS. Assume Cgate is 10**-14 F for a 2 x 2 transistor. 24 POINTS TO HERE

A	B	C	PU or PD?	time
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

4) (10)Design NAND(A,B) such that the following pullup and pulldown times are achieved. For AB=00, the pullup time constant is 0.1ns. For AB= 11, the pulldown time constant is 1.0 ns. Let the load be 100fF. Assume all channel lengths are 2. 34 POINTS TO HERE

5) (20) Realize (means draw the circuit) the function F=(AB+C) using a fully-complementary, fully switched (as contrasted with ratioed) CMOS circuit. Show the transistor level schematic. The signal, F, should be true at the output of the circuit. Use a design style that DOES require the use of an inverter, so that F appears true at the output. Assume all input variables and their complements are available. 54 POINTS TO HERE

6) (10) Realize the function F=AB using one domino stage showing all transistors. Indicate the precharge and evaluate transistors by labeling them. Also, label the clock. Be sure the output is a “valid” domino signal that can be fed to other NMOS logic, and that signal should be “true”, not “false”. 64 POINTS TO HERE

7) (15) Sketch a CMOS transistor embodiment of a full bit of dynamic shift that uses two dynamic half-bit shifters. Show the use of clocks phi1 and phi2 (not clock and clock_bar) as applied to the **FULLY-complementary CMOS switches**. You do not need to use transistors for the inverters, as it’s the switches I am interested in seeing done correctly. 79 POINTS TO HERE

8. What transistor is normally in the resistive region when an inverter is pulling down? (4) 83

9. What is the approximate resistance from a to b if the wire's sheet resistance is 100 ohms? (8) 91



10. A 1x inverter drives N-1 more inverters to form an optimal buffer chain to drive the final load of $3e^4$. What is N? (5) 96 to here

11. What is velocity saturation in short channel MOS transistors? Address both the origin of the effect and also whether the transistor still behaves as a square-law device. (4)